Chemistry Topic 2 — Matter, Solutions and Purification

Sources baked in: NCERT Class 9 & 11 (Matter in Our Surroundings; Is Matter Around Us Pure?; Solutions; Surface Chemistry), Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Questions on colloid types (milk as emulsion, smoke as aerosol), sublimation compounds, colligative applications (antifreeze, salting icy roads, pressure cookers), alum coagulation, and separation methods recur consistently across UPPCS, UKPCS, BPSC, and IAS Prelims. **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2025–2026	**Global Desalination & RO Tech**	Mega RO desalination plants in coastal India (Gujarat, Tamil Nadu) utilize **Reverse Osmosis** through thin-film composite polyamide semi-permeable membranes applying hydrostatic pressure $> \Pi$ (osmotic pressure, ~27–30 atm for seawater).
2024	**UPPCS Prelims 2024**	Direct question on **Colloidal Classification of Milk**: Identified strictly as an **emulsion** (liquid butterfat droplets dispersed in aqueous liquid phase stabilized by casein protein). Foams, sols, and aerosols were incorrect distractors.
2024	**Noble Prize & Quantum Condensates**	Bose-Einstein Condensate (BEC) research in microgravity aboard the ISS Cold Atom Lab (CAL): Atoms cooled to nanokelvin temperatures collapse into the lowest quantum ground state, acting as a single giant "super-atom" (predicted by Satyendra Nath Bose & Albert Einstein, 1924–25; realized by Cornell, Wieman & Ketterle, 1995).
2023	**UPPCS / UKPCS Trends**	Frequent testing of **Hardy-Schulze Rule**: Coagulation power of electrolytes increases exponentially with the valence of the active ion ($Al^{3+} > Mg^{2+} > Na^{+}$ for negative colloidal sols like clay/muddy water). Explains why potash alum ($K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$) rapidly purifies municipal water supplies.
2022	**UPPCS Prelims 2022**	Classic trap on **Physical vs. Chemical Change**: Curdling of milk and rusting of iron are irreversible chemical changes, whereas dissolution of sugar in water and melting of wax are physical changes. Burning of a candle involves **both** physical (melting of wax) and chemical (combustion of hydrocarbons into $CO_2 + H_2O$) changes.

Detailed Theory Notes

2.1 The Five States of Matter & Phase Transitions

1. Fundamental States: Kinetic Molecular Theory

Matter is characterized by two competing influences: **intermolecular attractive forces** (which draw molecules together into ordered arrays) and **thermal kinetic energy** (which drives molecules apart into

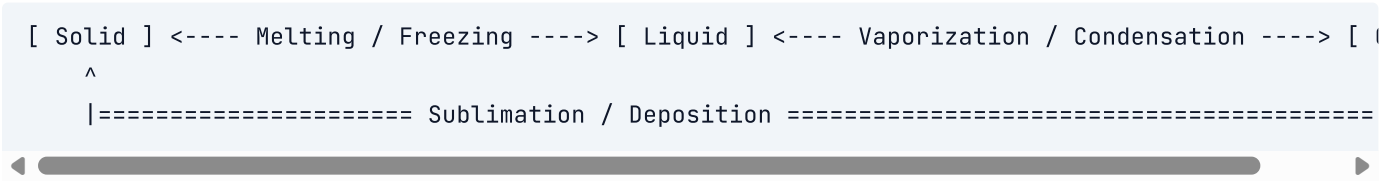
chaotic motion). - **Solid State**: Thermal kinetic energy is far weaker than intermolecular lattice attractions. Particles are packed into regular crystalline lattices or rigid amorphous frameworks. They exhibit minimum entropy (S), fixed geometry, and incompressibility. - **Liquid State**: Thermal kinetic energy is roughly comparable to intermolecular attractions. Molecules possess sufficient energy to glide past one another while remaining within attractive interaction fields. Liquids exhibit fluidity, definite volume, surface tension, and viscosity. - **Gaseous State**: Thermal kinetic energy completely overcomes intermolecular attractions. Molecules travel at hundreds of meters per second in straight lines between random elastic collisions. Intermolecular space is massive relative to molecular diameters.

2. Plasma: The Fourth State of Matter

- When a neutral gas is subjected to extreme thermal energy (thousands of degrees Celsius) or immense electrical discharges, valence electrons are stripped from atomic nuclei. The resulting state is an electrically quasi-neutral soup of **free electrons, positive ions, and excited neutral atoms** termed **plasma** (coined by Irving Langmuir in 1928).
- **Properties**: Highly electrically conductive, responsive to external electric and magnetic fields, emits intense radiant photons.
- **Occurrences**: Sun and stars (maintained by nuclear fusion), interstellar nebulae, auroras (Borealis/Australis), lightning bolts, neon advertising signs, fluorescent tubes, and experimental thermonuclear tokamak reactors (ITER).

3. Bose-Einstein Condensate (BEC): The Fifth State

- **Theoretical Origin**: Predicted theoretically in 1924–1925 by Indian physicist **Satyendra Nath Bose** (deriving quantum photon statistics) and generalized by **Albert Einstein** for integer-spin atoms (bosons).
- **Mechanism**: When a dilute gas of bosonic atoms (such as Rubidium-87, Sodium-23, or Caesium-133) is cooled using laser cooling and magnetic evaporative cooling to within a few nanokelvins above absolute zero ($< 100\ \text{nK}$), their de Broglie wavelengths ($\lambda = h/p$) expand until they overlap.
- **Result**: Individual atomic identities merge into a single macroscopic quantum wave packet—a "super-atom" occupying the zero-momentum quantum ground state.
- **Experimental Proof**: Created experimentally in 1995 by **Eric Cornell and Carl Wieman** at JILA (Boulder, Colorado) and independently by **Wolfgang Ketterle** at MIT (awarded the Nobel Prize in Physics, 2001).



4. Latent Heat & Phase Transition Thermodynamics

- **Latent Heat of Fusion (L_f)**: The quantity of heat energy required to convert $1\ \text{kg}$ of a solid into a liquid at its melting point under $1\ \text{atm}$ of pressure without any increase in temperature. For ice, $L_f \approx 3.34 \times 10^5\ \text{J/kg}$ ($80\ \text{cal/g}$).
- **Latent Heat of Vaporization (L_v)**: The quantity of heat required to convert $1\ \text{kg}$ of a liquid into vapor at its boiling point under $1\ \text{atm}$ pressure without any temperature change. For water, $L_v \approx 2.26 \times 10^6\ \text{J/kg}$ ($540\ \text{cal/g}$).

- **Must-Score**: Steam at 100°C causes much more severe burns than boiling water at 100°C because each gram of steam contains an additional **540 calories (2260 J) of latent heat of vaporization**, which is released directly onto the skin upon condensation.

5. Triple Point and Critical Point

- **Triple Point**: The unique thermodynamic invariant where solid, liquid, and gaseous phases coexist in dynamic equilibrium. For pure water:

$$\text{Temperature} = 0.01^\circ\text{C} = 273.16\ \text{K}, \quad \text{Pressure} = 4.58\ \text{mmHg} = 611.65\ \text{Pa} = 0.00603\ \text{atm}$$

- **Critical Point**: The termination point of a phase equilibrium curve. Above the critical temperature (T_c) and critical pressure (P_c), the boundary between liquid and gas vanishes, yielding a **Supercritical Fluid** possessing the density of a liquid but the diffusivity and low viscosity of a gas (e.g., supercritical CO_2 used industrially for decaffeinating coffee beans).

2.2 Physical vs. Chemical Transformations

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                          ┌── Physical Change: No new substance; reversible; mass/bonds preserve
Changes in Matter ────────┤   (e.g., Melting of ice, glowing of bulb, dissolving sugar)
                          └── Chemical Change: New substances; chemical bonds broken/made; irrev
                              (e.g., Rusting of iron, curdling of milk, digestion of food)
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1. Defining Criteria

- **Physical Changes**: Involve alterations in physical properties (state of matter, density, color appearance, shape) while leaving the intrinsic chemical formula and molecular structure unaltered. Intermolecular forces are broken or formed, but intramolecular covalent/ionic bonds remain intact. Generally reversible by reversing thermodynamic conditions.
- **Chemical Changes**: Involve the rupture of existing chemical bonds and the formation of new chemical bonds, resulting in new chemical species with distinct electronic configurations, chemical properties, and stoichiometry. Generally irreversible by simple physical manipulation.

2. Key High-Yield Phenomena Analyzed

1. **Rusting of Iron**:

$$4Fe + 3O_2 + 2xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O\ (\text{Hydrated Ferric Oxide})$$

   - **Nature**: Pure chemical change (electrochemical redox reaction). - **Mass Effect**: The weight of an iron article **increases upon rusting** because the iron atoms chemically bind external oxygen and water molecules from the atmosphere into the crystalline rust matrix. - **Requirements**: Requires the simultaneous presence of **both oxygen and moisture**. Rusting is accelerated in saline/marine environments (electrolytes enhance ionic conductivity).

2. **Burning of a Candle**: - Paraffin wax consists of higher solid alkanes ($C_{20}H_{42}$ to $C_{30}H_{62}$). - **Physical Aspect**: Solid wax melts near the wick into liquid wax, rises by capillary action, and vaporizes into wax vapor (changes of state without chemical transformation). - **Chemical Aspect**: Wax vapor reacts with atmospheric oxygen at flame temperatures to produce carbon dioxide, carbon monoxide, water

vapor, and luminous carbon soot ($C_nH_{2n+2} + O_2 \rightarrow CO_2 + H_2O + \text{heat} + \text{light}$). Thus, burning of a candle represents **both a physical and a chemical change**.

3. **Curdling of Milk**: - Lactic acid bacteria (*Lactobacillus*) ferment lactose sugar ($C_{12}H_{22}O_{11}$) into lactic acid ($CH_3CH(OH)COOH$). - The drop in pH below 4.6 reaches the isoelectric point of **casein protein**, causing casein micelles to denature, coagulate, and precipitate out into curds. This is an **irreversible chemical change**.
4. **Dissolution of Common Salt or Sugar in Water**: - $NaCl\,(\text{solid}) + H_2O \rightarrow Na^+(\text{aq}) + Cl^-(\text{aq})$. - Solute ions are hydrated by polar water molecules. When water is evaporated, pure crystalline $NaCl$ is completely recovered with identical crystal structure and mass. Hence, dissolution of salt/sugar is a **physical change**.

2.3 Solutions, Solubility & Colligative Properties

1. Solutions and Henry's Law

A **True Solution** is a homogeneous mixture of two or more chemically non-reacting substances whose composition can be varied within certain limits. - **Henry's Law**: At constant temperature, the mass (m) or solubility of a gas dissolved in a given volume of liquid is directly proportional to the partial pressure (P) of the gas in equilibrium with the liquid:

$$P = K_H \cdot X$$

where K_H is Henry's law constant and X is the mole fraction of gas in solution. - **Key Applications**: 1. *Carbonated Beverages*: Bottles are sealed under high hydrostatic pressure of CO_2 gas. Opening the cap suddenly lowers the pressure, causing solubility to collapse and triggering rapid bubbling (effervescence). 2. *Scuba Diving & Decompression Sickness ("Bends")*: At high ocean depths, high hydrostatic pressure forces large quantities of atmospheric nitrogen to dissolve in the diver's blood. When the diver ascends too rapidly, the sudden drop in pressure causes dissolved N_2 to come out of solution as gas bubbles in the blood vessels, blocking capillaries and causing excruciating pain or death. To prevent this, deep-sea diving gas cylinders are filled with oxygen diluted with **Helium** ($11.7\%\ \text{He}, 56.2\%\ \text{N}_2, 32.1\%\ \text{O}_2$) because helium exhibits exceptionally low blood solubility. 3. *Anoxia at Mountain Altitudes*: At high mountain elevations, atmospheric partial pressure of O_2 is significantly lower than at sea level. The concentration of dissolved oxygen in the blood and brain tissues drops, causing cognitive weakness, fatigue, and impaired judgment, a condition known as **anoxia**.

2. Colligative Properties & Thermodynamic Principles

Colligative properties depend exclusively on the **ratio of the number of solute particles to the number of solvent molecules**, not on the chemical identity of the solute.

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                              ┌── 1. Relative Lowering of Vapor Pressure (Raoult's Law)
                              ├── 2. Elevation of Boiling Point (ΔTb = i · Kb · m)
Colligative Properties ──────┤
                              ├── 3. Depression of Freezing Point (ΔTf = i · Kf · m)
                              └── 4. Osmotic Pressure (Π = i · C · R · T)
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1. **Relative Lowering of Vapor Pressure (Raoult's Law)**: When a non-volatile solute is added to a solvent, solute particles occupy a fraction of the surface liquid layer, decreasing the escape rate of

solvent molecules. The relative lowering of vapor pressure equals the mole fraction of the solute:

$$\frac{P^\circ - P_s}{P^\circ} = X_{\text{solute}}$$

2. **Elevation of Boiling Point**: Boiling occurs when the liquid's vapor pressure equals external atmospheric pressure. Because adding a non-volatile solute lowers vapor pressure, the solution must be heated to a higher temperature to reach atmospheric pressure:

$$\Delta T_b = T_b - T_b^\circ = i \cdot K_b \cdot m$$

where K_b is the ebullioscopic constant (for water, $K_b = 0.52\ \text{K} \cdot \text{kg/mol}$) and i is the van 't Hoff factor. - *Cooking Application*: Adding table salt ($NaCl$) to boiling water elevates its boiling point slightly, providing a hotter cooking bath. - *Pressure Cooker*: Increasing the chamber pressure to $2\ \text{atm}$ raises the boiling point of water to $\sim 120^\circ\text{C}$, cooking food far faster because cooking rate doubles approximately every 10°C rise in temperature.

3. **Depression of Freezing Point**: A liquid freezes when its liquid vapor pressure equals the vapor pressure of its solid phase. Lowering the vapor pressure lowers the temperature at which solid and liquid coexist:

$$\Delta T_f = T_f^\circ - T_f = i \cdot K_f \cdot m$$

where K_f is the cryoscopic constant (for water, $K_f = 1.86\ \text{K} \cdot \text{kg/mol}$). - *Automobile Radiators*: Adding **ethylene glycol** ($C_2H_6O_2$) lowers the freezing point of radiator water to -20°C to -40°C, preventing the cooling liquid from freezing and cracking the engine block in sub-zero winters. - *Clearing Snow from Roads*: Scattering rock salt ($NaCl$) or calcium chloride ($CaCl_2$) over icy roads depresses the freezing point of water below the ambient road temperature, causing the ice to melt into brine.

4. **Osmotic Pressure (Π) & Reverse Osmosis**: The hydrostatic pressure required to prevent the osmotic migration of pure solvent through a semi-permeable membrane:

$$\Pi = i \cdot C \cdot R \cdot T = i \cdot \left(\frac{n}{V}\right) RT$$

- *Reverse Osmosis (RO)*: If external hydrostatic pressure $P > \Pi$ is applied to the concentrated saline side, solvent molecules move against the chemical potential gradient—from concentrated solution to pure solvent. Synthetic semi-permeable membranes (cellulose triacetate, cross-linked aromatic polyamide) reject dissolved salts ($Na^+, Cl^-, Mg^{2+}, SO_4^{2-}$) and pass purified potable water.

2.4 Colloidal State & Surface Chemistry

1. Particle Size Spectrum

- **True Solution**: Particle diameter $< 1\ \text{nm}$ ($< 10\ \text{\AA}$). Homogeneous, completely transparent, passes through filter paper and animal membranes, does not scatter light, does not settle.
- **Colloidal Solution**: Particle diameter between $1\ \text{nm}$ **and** $1000\ \text{nm}$ **(**$10^{-9}\ \text{m}$ **to** $10^{-6}\ \text{m}$**)**. Heterogeneous, exhibits Tyndall effect, particles pass through ordinary filter paper but are retained by parchment or ultrafilter membranes.

- **Suspension**: Particle diameter $> 1000\ \text{nm}$ ($> 1\ \mu\text{m}$). Heterogeneous, opaque, particles visible to the naked eye or optical microscope, retained by filter paper, settles spontaneously under gravity.

2. Key Colloidal Properties

1. **Optical Property — Tyndall Effect**: - First observed by Michael Faraday and investigated systematically by John Tyndall (1869). - When an intense light beam passes through a colloidal sol in a darkened room, the path of the beam is illuminated as a bright cone (Tyndall cone). - *Mechanism*: Colloidal particle diameters are comparable to the wavelength of visible light ($400–700\ \text{nm}$), scattering light in all directions. - *Real-world manifestations*: Blue color of the sky (Rayleigh scattering by atmospheric colloids), sunlight streaming through dense forest canopies, projector beam visible in a smoky movie theater.
2. **Kinetic Property — Brownian Motion**: - First observed by Scottish botanist Robert Brown (1827) under a microscope with pollen grains suspended in water; explained theoretically by Albert Einstein (1905). - Continuous, chaotic, random zig-zag motion of colloidal particles caused by unequal, asymmetric kinetic collisions delivered by the molecules of the dispersion medium. - *Significance*: Brownian motion acts as a stirring mechanism that counteracts gravitational sedimentation, conferring thermodynamic stability on colloidal sols.
3. **Electrical Property — Zeta Potential & Electrophoresis**: - Colloidal particles always carry a net positive or negative electric surface charge due to preferential adsorption of common ions or dissociation of surface groups. - **Electrophoresis**: When an external electric field is applied across a colloidal sol via two electrodes, colloidal particles migrate toward the oppositely charged electrode (positively charged particles migrate to cathode $\rightarrow$ cataphoresis; negatively charged to anode $\rightarrow$ anaphoresis). - Confirms that colloidal stability is maintained by electrostatic repulsion between similarly charged particles.
4. **Coagulation (Flocculation) & Hardy-Schulze Rule**: - Neutralizing the surface electrical charge of colloidal particles allows van der Waals attractive forces to pull them together into macroscopic aggregates that precipitate out of suspension. - **Hardy-Schulze Law**:
    1. The effective coagulating ion is the ion carrying a charge **opposite** to that of the colloidal particles.
    2. The coagulating power of an electrolyte ion increases steeply with its valence:

$$\text{Coagulating Power} \propto (\text{Valence})^4$$

    - For negatively charged sols (e.g., As_2S_3 sol, clay in muddy water):

$$\text{Coagulation Power: } Al^{3+} \gg Ba^{2+} > Na^{+} \quad (\text{Ratio } \approx 750 : 30 : 1)$$

    - For positively charged sols (e.g., $Fe(OH)_3$ sol):

$$\text{Coagulation Power: } [Fe(CN)_6]^{4-} > PO_4^{3-} > SO_4^{2-} > Cl^{-}$$

3. Real-World Applications of Coagulation

- **Alum in Water Purification**: Potash alum ($K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$) releases trivalent Al^{3+} ions in water. These Al^{3+} ions neutralize the negative surface charges on suspended clay and silt colloids, coagulating them into heavy flocculent precipitates that settle rapidly to the bottom of the settling tank.

- **Delta Formation at River Mouths**: River water is a colloidal suspension of negatively charged clay particles. Seawater contains high concentrations of electrolytes (Na^+, Mg^{2+}, Ca^{2+}). When river water encounters the sea, the multivalent cations neutralize the clay colloids, causing immediate coagulation and massive silt deposition that builds delta landforms over centuries.
- **Hemostatic Action of Alum / Styptic Pencils**: Blood is a negatively charged colloidal sol of albumin, hemoglobin, and platelets. Applying alum (Al^{3+}) or ferric chloride ($FeCl_3$, Fe^{3+}) rapidly coagulates blood proteins, forming a protective clot that seals the wound and halts bleeding.

Bilingual Terminology -- Matter, Solutions and Purification

English Term	Hindi Term	Key Anchor
Matter (पदार्थ)	पदार्थ	Anything that has mass and occupies space; solid, liquid, gas, plasma
Solid (ठोस)	ठोस	Fixed shape and volume; strong intermolecular forces
Liquid (तरल)	तरल / द्रव	Fixed volume but takes container's shape; moderate intermolecular forces
Gas (गैस)	गैस / वायु	No fixed shape or volume; weak intermolecular forces
Plasma (प्लाज्मा)	प्लाज्मा	4th state; ionized gas; found in stars, lightning, neon signs
Mixture (मिश्रण)	मिश्रण	Two or more substances mixed; no new substance formed
Solution (घोल / विलयन)	विलयन	Homogeneous mixture; solute dissolved in solvent
Solute (विलेय)	विलेय	Substance that dissolves in solvent
Solvent (विलायक)	विलायक	Substance that dissolves the solute; water = "universal solvent"
Colloid (कोलाइड)	कोलाइड	Heterogeneous mix; particle size 1-1000 nm; e.g., milk, fog
Suspension (निलम्बन)	निलम्बन	Heterogeneous; particles > 1000 nm; settle on standing; e.g., muddy water
Tyndall Effect (टिंडल प्रभाव)	टिंडल प्रभाव	Scattering of light by colloidal particles; sky in morning, beam in dusty room
Filtration (निस्यंदन)	निस्यंदन	Separates insoluble particles from liquid using filter paper
Distillation (आसवन)	आसवन	Separation by vaporisation and condensation; purifies liquids
Chromatography (वर्णलेखन)	वर्णलेखन	Separation based on different rates of movement through a medium
Crystallisation (क्रिस्टलीकरण)	क्रिस्टलीकरण	Purification by forming pure crystals from solution; for salt, sugar, alum

One-Liner Revision -- Matter, Solutions and Purification

#	Fast Fact
1	States of matter: Solid, Liquid, Gas, Plasma (4th state in stars/lightning)
2	Sublimation = solid directly to gas (dry ice/CO2, iodine, naphthalene, camphor)
3	Deposition = gas directly to solid (frost formation)
4	Water: boiling point = 100 degC; freezing = 0 degC; maximum density at **4 degC**
5	Universal solvent = **Water** (H2O)
6	Solution components: solute (minor, dissolved) + solvent (major, dissolving)
7	Mixture types: homogeneous (uniform) = solution; heterogeneous (non-uniform) = suspension
8	Colloid particle size = 1-1000 nm; shows Tyndall effect; e.g., milk, fog, smoke
9	Tyndall effect = scattering of light by colloidal particles (explains why beam visible in dusty room)
10	Suspension = particles > 1000 nm; settles on standing; filtered by filter paper
11	Distillation = separates liquids with different boiling points
12	Fractional distillation = separates crude oil into petrol, diesel, kerosene fractions
13	Chromatography = separates pigments, amino acids (paper, column, gas, HPLC)
14	Centrifugation = separates components by density using centrifugal force; cream from milk
15	Crystallisation = purification method; produces pure crystals; for alum, salt
16	Miscible liquids = dissolve in each other in all proportions; e.g., alcohol + water
17	Immiscible liquids = do not mix; separated by separating funnel; e.g., oil + water
18	Henry's Law: solubility of gas in liquid increases with pressure (soda carbonation)
19	Molarity (M) = moles of solute per litre of solution
20	Molality (m) = moles of solute per kg of solvent
21	Saturated solution = maximum solute dissolved at given temp; undissolved solute remains
22	Supersaturated = more solute than normally possible; unstable; seeding causes crystallisation
23	Hard water = contains Ca2+ and Mg2+ salts; does not form lather easily
24	Temporary hardness = Ca/Mg bicarbonates; removed by boiling or lime
25	Permanent hardness = Ca/Mg sulphates/chlorides; removed by washing soda or ion exchange
26	Alloy = mixture of metals (or metal + non-metal); e.g., brass = Cu+Zn, steel = Fe+C
27	Amalgam = alloy with mercury; dental fillings (Ag-Sn-Hg amalgam)
28	Electrolysis = chemical decomposition using electricity; used to refine metals, electroplate

#	Fast Fact
29	Osmosis = water moves from low to high solute concentration through semipermeable membrane
30	Reverse osmosis (RO) = applied pressure forces water against osmosis; water purification